

Superstore Sales Dataset Analysis Report

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Overview

This report provides a comprehensive analysis of sales performance, product distribution, and shipment activity for Super Store based on historical data from 2015 to 2018. Using Power BI dashboards, key trends and insights were identified across different customer segments, categories, and geographical regions. The analysis aims to assist stakeholders in understanding sales dynamics, identifying top-performing products, optimizing shipping strategies, and guiding future decision-making processes.

Project Description

This project aims to analyze a store sales dataset using Power BI to extract valuable insights that can support decision-making processes. The dataset will undergo data cleaning, transformation, and visualization to ensure accuracy and clarity. The final output will be an interactive Power BI dashboard that presents key findings effectively. The project will be divided into four main phases: data cleaning, analysis questions formulation, dashboard creation, final report and presentation.

Objective

The main objective of this project is to analyze Super Store data into three main subjects:

1. Sales Performance analysis

- What is the total sales value over the years?
- How have sales grown or declined year-over-year?
- Which customer segments contribute the most to revenue?
- Who are the top 5 customers by total sales?
- Which customer segments (Consumer, Corporate, Home Office) perform best?

2. Top Product Overview.

- What are the best-selling products?
- How many products were sold this year versus previous periods?
- Which product categories and sub-categories dominate the sales?
- How does product performance vary by region?
- What is the average order value and product count per customer?

3. Shipment Modes insights.

- What are the best-selling products?
- Which cities and states generate the highest sales?
- How is product distribution spread across different regions?
- What are the most frequently used shipping methods?
- How does shipping mode affect sales?
- What is the average delivery time and how does it affect sales volume?
- How many days does it typically take to deliver products?

Data Sources & Preparation

Data Inspection Summary

The dataset "Superstore Sales Dataset contains the following key columns:

- **Order Info:** Order ID, Order Date
- **Customer Info:** Customer ID, Customer Name, Segment,
- **Location Info:** Country, City, State, Postal Code, Region
- **Ship Info :** Ship Date, Ship Mode , Shipping Duration (to be calculated)
- **Product Info:** Product ID, Category, Sub-Category, Product Name
- **Sales Info:** Sales

Raw Data Source:

CSV File: Superstore Sales Dataset.csv

Data Cleaning Steps

1. Initial Inspection in Excel

- **Reviewed null values, formats, and duplicates:** by Reviewing date values in order date & ship date column we found that some values were not in the same date format so we used excel to unify the format to ensure a successful data load in Power Bi.

2. Loading Data into Power BI (Power Query Editor)

- changed data types, created new columns for date intelligence and shipping duration.

Data Modeling & Dax Calculations

Tables Created:

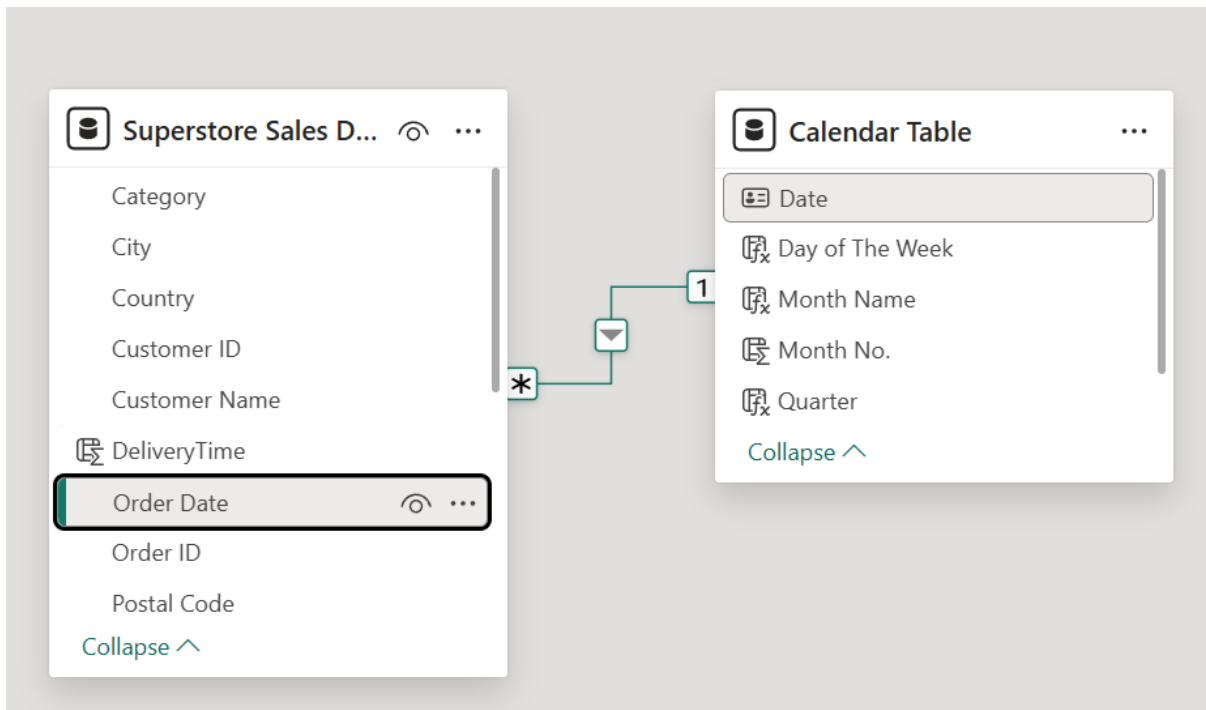
- Fact Table.
- Dimension Tables: None
- Calendar Table

Date ▾	Month No. ▾	Month Name ▾	Quarter ▾	Day of The Week ▾	Year ▾
03/01/2015	1	January	Q1	Saturday	2015
04/01/2015	1	January	Q1	Sunday	2015
05/01/2015	1	January	Q1	Monday	2015
06/01/2015	1	January	Q1	Tuesday	2015
07/01/2015	1	January	Q1	Wednesday	2015
08/01/2015	1	January	Q1	Thursday	2015
09/01/2015	1	January	Q1	Friday	2015
10/01/2015	1	January	Q1	Saturday	2015
11/01/2015	1	January	Q1	Sunday	2015
12/01/2015	1	January	Q1	Monday	2015
13/01/2015	1	January	Q1	Tuesday	2015
14/01/2015	1	January	Q1	Wednesday	2015
15/01/2015	1	January	Q1	Thursday	2015
16/01/2015	1	January	Q1	Friday	2015
17/01/2015	1	January	Q1	Saturday	2015
18/01/2015	1	January	Q1	Sunday	2015

Model Relationships:

- Calendar ↔ Superstore Sales Dataset Table (Order Date):

One to Many Relationship



DAX Measures Examples:

```
PYTD sales = CALCULATE(SUM('Superstore Sales Dataset'[Sales]), SAMEPERIODLASTYEAR('calender table'[Date]))
```

```
Product_Count = COUNT('Superstore Sales Dataset'[Product ID])
```

```
sales diff color = IF([sales diff]> 0 , "green", "red")
```

```
1 Min order value = MINX(  
2     SUMMARIZE(  
3         'Superstore Sales Dataset',  
4         'Superstore Sales Dataset'[Order ID],  
5         "إجمالي قيمة الفاتورة", SUM('Superstore Sales Dataset'[Sales])  
6     ),  
7     [إجمالي قيمة الفاتورة]  
8 )
```

```

1 Max order value = MAXX(
2   SUMMARIZE(
3     'Superstore Sales Dataset',
4     'Superstore Sales Dataset'[Order ID],
5     "إجمالي قيمة الفاتورة", SUM('Superstore Sales Dataset'[Sales])
6   ),
7   [إجمالي قيمة الفاتورة]
8 )

```

```
Growth color = IF([Sales Growth]< 0, "Low", "High")
```

```
sales diff = [YTD total sales]- [PYTD sales]
```

```

1 AverageOrdersValue =
2 DIVIDE(
3   SUM('Superstore Sales Dataset'[Sales]),
4   DISTINCTCOUNT('Superstore Sales Dataset'[Order ID])
5 )

```

```
Avg ship days = SUM('Superstore Sales Dataset'[Diff date])/ COUNTROWS('Superstore Sales Dataset')
```

```
Total Sales = SUM('Superstore Sales Dataset'[Sales])
```

✓

Analysis Results

1. Sales Page

Highlights:

- **Total Sales:** \$2.3M with a -53% growth compared to previous periods.
- **Year-To-Date (YTD) Sales:** \$722.1K.
- **Average Order Value (AOV):** \$459.5 with an average sale of \$230.8.

Visuals Explained:

- **Sales Per Date:** A bar chart tracking total sales annually from 2015 to 2018 with a trend line.
- **Category by Sales:** Shows sales distribution across Consumer, Corporate, and Home Office segments.
- **Sales by Geographical Area:** Bar chart listing top-performing cities (e.g., New York City, Los Angeles).
- **Top 5 Customer Sales:** Ranked table highlighting customers with the highest sales totals.
- **Top 10 Products by Sales:** Bar chart of best-selling products, led by "Canon image..." at 62K.
- **Sales by Segment (Pie Chart):** Breakdown showing Consumer as the top segment (over 50%).



2. Products Page

Highlights:

- **Total Products Sold:** 9,800.
- **YTD Product Sales:** 3,258.
- **Average Products Per Customer (APPC):** 2.3.
- **Average Sales per Product:** \$1.2K.

Visuals Explained:

- **Product Count by Segment:** Consumer leads with 5.1K products sold.
- **Top Products Sold:** "Staple envelopes" and "Staples" dominate the chart.
- **Product Count by Region:** West region leads with 3.1K units.
- **Sankey Diagram (Product Flow):** Visualizes flow from region to product name, showing distribution across regions and categories.
- **Product Count by Category:** Office Supplies leads with 5.91K units.



3. Shipment Page

Highlights:

- **Total Shipments Count:** 9,800.
- **Average Shipment Time:** 3.96 days.
- **Most Common Ship Mode:** Standard Class (5.9K shipments).

Visuals Explained:

- **Product Count by Ship Mode:** Bar chart showing shipment count by mode.
- **Sum of Sales by Ship Mode (Pie Chart):** Standard Class dominates with 59.29% of sales.
- **Product Count by Region:** Mirrors the Products page with similar regional distribution.
- **Map of Product Count by State:** Bubble map showing sales intensity across US states.
- **Sales by Delivery Time (Line Chart):** Plots sales based on delivery duration from 0 to 7+ days.



Key Findings

- Sales are highest in the "Consumer" segment (50.7%).
- "Canon Image" products are top-selling.
- "West" region has most activity.
- Standard shipping correlates with higher Sales.

- **Low performance** in cities like Detroit – potential areas for **growth**.

Key Questions Answered:

1. What is the total sales value over the years?

- Sales have grown from \$459.4K in 2016 to \$722.1K in 2018, peaking in 2018. The total cumulative sales value shown is \$2.3M.

2. Which customer segments contribute the most to revenue?

- The Consumer segment leads significantly with \$1.15M in sales, accounting for over 50% of the total.

3. Who are the top 5 customers by total sales?

The top customers are:

- Sean Miller – \$25,043.05
- Tamara Chand – \$19,052.22
- Raymond Buch – \$15,117.34
- Tom Ashbrook – \$14,595.62
- Adrian Barton – \$14,473.57

4. What are the best-selling products?

- The leading product by sales is **Canon image 2200** at \$62K, followed by **Fellowes PB500 Electric Punch Plastic** (\$27K) and **Cisco TelePresence System EX90** (\$23K).

5. How does product performance vary by region?

- The West region leads in product sales with 3.1K units, followed by East (2.8K), Central (2.3K), and South (1.6K).

6. Which cities and states generate the highest sales?

- New York City is the top-performing city with \$0.25M in sales. Other top cities include Los Angeles and Seattle.

7. What are the most frequently used shipping methods?

- Standard Class dominates with 5.9K shipments, accounting for 59.29% of total sales.

8. What is the average delivery time and how does it affect sales?

- The average shipment time is 3.96 days. Sales tend to peak on delivery days 3, 4, and 5, especially on day 4 with \$621.54K in sales.

9. How many products were sold this year versus previous periods?

- The Year-To-Date (YTD) product sales are 3,258, from a total of 9,800 products sold overall.

10. What is the average order and product count per customer?

- The Average Order Value (AOV) is \$459.5
- The Average Products Per Customer (APPC) is 2.3, with average sales per product of \$1.2K.

Recommendations to Improve Sales

- **Focus on the Consumer Segment:**
Since the Consumer segment accounts for over 50% of total sales, enhance marketing campaigns tailored to this segment and introduce loyalty programs.
- **Leverage High-Performing Products:**
Promote top-selling products like "Canon image 2200" and " Fellowes PB500 Electric Punch Plastic " more aggressively, especially in high-performing regions.
- **Improve Sales in Underperforming Regions:**
The South region shows the lowest product sales (1.6K). Investigate underlying causes and increase marketing or availability in those areas.
- **Enhance Shipping Speed and Options:**
With an average delivery time of 3.96 days, consider introducing faster shipping methods or improving current logistics to meet customer expectations.
- **Promote to Top Cities:**
Maximize potential in cities like **New York and Los Angeles** by increasing advertising presence and exploring new physical or digital retail points.
- **Optimize Product Mix by Region:**
Tailor inventory and product availability based on regional preferences identified in the Sankey and regional sales data.
- **Encourage Bulk Buying:**
With an average of 2.3 products per customer, offer bulk purchase discounts to increase order size.
- **Target High-Value Customers:**
Engage top spenders like Sean Miller and Tamara Chand with personalized deals, early access to promotions, or VIP programs to retain and grow their accounts.
- **Improve shipping in underperforming areas.**
 - Expand to the South region to improve regional balance.
 - Invest in faster delivery methods to drive satisfaction and repeat purchases.